

10. The primary function of a gearbox in an automobile is to:
 A) Increase engine power B) Reduce fuel consumption
 C) Provide different torque and speed ratios D) Control braking force
11. Which type of gearbox is used in automatic transmission vehicles?
 A) Sliding mesh gearbox B) Epicyclic gearbox
 C) Constant mesh gearbox D) Synchromesh gearbox
12. The differential is located between:
 A) The engine and the clutch B) The rear wheels in a rear-wheel drive vehicle
 C) The gearbox and the driveshaft D) The front wheels in all vehicles
13. What type of suspension is commonly used in the front wheels of modern passenger cars?
 A) MacPherson strut suspension B) Leaf spring suspension
 C) Rigid axle suspension D) Multi-link suspension
14. Which of the following is NOT a type of suspension system?
 A) Independent suspension B) Semi-independent suspension
 C) Rigid axle suspension D) Torque converter suspension
15. The purpose of caster angle in steering geometry is to:
 A) Reduce tire wear B) Increase the turning radius
 C) Improve straight-line stability D) Improve fuel efficiency
16. In modern cars, which type of brake is commonly used in the front wheels?
 A) Drum brake B) Disc brake
 C) Band brake D) Electromagnetic brake
17. The primary function of a tyre in an automobile is to:
 A) Increase engine power B) Reduce vehicle weight
 C) Provide traction and absorb shocks D) Control air resistance
18. Which of the following technologies is commonly used in vehicles to meet stringent emission standards?
 A) Turbocharging B) Exhaust Gas Recirculation (EGR)
 C) Carburetors D) Diesel fuel only
19. The term "aspect ratio" in a tyre refers to:
 A) The width of the tyre B) The height of the sidewall as a percentage of the tyre width
 C) The diameter of the wheel D) The weight of the tyre
20. What is the purpose of the Anti-lock Braking System (ABS)?
 A) To increase vehicle speed B) To reduce fuel consumption
 C) To prevent wheel lock-up during braking D) To control tire pressure

PART - B: DESCRIPTIVE ANSWER QUESTIONS

Unit – I		Marks	BT*	CO*	PO*
1.	a) Explain with neat sketch working of a four stroke petrol engine.	8	L*2	1	1
	b) What is the function of lubricating system in an engine? Explain with sketch working of wet sump lubrication system.	8	L2	2	1
2.	a) What is cylinder liner? Explain wet and dry liner with sketch.	8	L2	1	1
	b) Explain construction and working of AC mechanical pump with sketch.	8	L2	2	1
3.	a) What is the necessity of cooling system in an engine? Explain working of thermo syphon system of water cooling with a neat diagram.	8	L2	1	1

- b) With a neat circuit diagram explain working of a four cylinder engine battery ignition system.

8 L2 2 1

Unit – II

4. a) Explain the working of a constant mesh gearbox.
b) With neat sketch explain the construction and working of differential.
5. a) Explain the following with a neat diagram:
i) Caster and camber ii) Toe-in and toe-out
b) Explain the working single-plate clutch with a neat diagram.
6. a) Explain the working of the torque converter with a neat diagram.
b) Explain the working of the Hotchkiss drive with a neat sketch.

8 L2 3 1

8 L2 4 1

8 L2 3 1

8 L2 4 1

8 L2 3 1

8 L2 4 1

Unit – III

7. a) Explain the working of air braking system with line diagram.
b) What is the purpose of EGR? Explain the working of EGR system.
8. a) Explain the working of hydraulic braking system.
b) Differentiate between tubeless tyres and tubed tyres with necessary sketches.

8 L2 5 1

8 L2 5 1

8 L2 5 1

8 L2 5 1

BT* Bloom's Taxonomy, L* Level; CO* Course Outcome; PO* Program Outcome
